

GATE DA Practice Set

GATE DA Practice Set — Distributions, Naive Bayes, Number Systems

Coverage: Standard Distributions, Normal Distribution, Exponential Distribution, Joint Distribution, Naive Bayes Classifier, Number Systems.

Note on sourcing: GATE DA has only run in 2024, 2025 and 2026, so genuinely verified PYQs on some of these exact sub-topics are limited. Every question below is labeled honestly: - **PYQ** = confirmed real GATE DA/DS&AI question (concept and numbers matched to the original; phrasing reworded) - **PYQ-PATTERN** = mirrors a real, well-known GATE-style question structure but is not a verified exact paper question - **ORIGINAL** = newly written, GATE DA-level
No year/question-number is stated unless it is reliably tied to a confirmed source.

1. Standard Distributions

Question 1

Type: MSQ Source: PYQ-PATTERN (based on GATE DA 2024 memory-based paper) Difficulty: Easy

Question: Consider the following statements about probability distributions: (S1) For a Poisson distribution, the mean and variance are equal. (S2) For a standard normal distribution, the mean is 0 and the variance is 1. Which of the statements is/are TRUE?

Options: (A) Only S1 (B) Only S2 (C) Both S1 and S2 (D) Neither S1 nor S2

Answer: (C)

Solution: Poisson(λ) has mean = variance = λ , so S1 is true. Standard normal is defined as $N(0,1)$, so S2 is true by definition.

Concept Tested: Basic properties of standard distributions

Question 2

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: X follows a Uniform distribution on $[2, 10]$. Find $\text{Var}(X)$.

Answer / Expected Answer: 5.33 (i.e., $16/3$)

Solution: For $U(a,b)$, $\text{Var}(X) = (b-a)^2/12 = 8^2/12 = 64/12 = 16/3 \approx 5.33$.

Concept Tested: Uniform distribution variance formula

Question 3

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: $X \sim \text{Binomial}(n=10, p=0.4)$. Find $\text{Var}(X)$.

Options: (A) 1.6 (B) 2.4 (C) 4.0 (D) 6.0

Answer: (B)

Solution: $\text{Var}(X) = np(1-p) = 10 \times 0.4 \times 0.6 = 2.4$.

Concept Tested: Binomial mean/variance

Question 4

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim \text{Poisson}(\lambda=3)$. Find $P(X=0)$ (round to 4 decimals).

Answer / Expected Answer: 0.0498

Solution: $P(X=0) = e^{-\lambda} = e^{-3} \approx 0.0498$.

Concept Tested: Poisson PMF

Question 5

Type: MCQ Source: PYQ-PATTERN Difficulty: Medium

Question: Which discrete distribution satisfies $P(X > m+n \mid X > m) = P(X > n)$ for non-negative integers m, n ?

Options: (A) Binomial (B) Poisson (C) Geometric (D) Uniform (discrete)

Answer: (C)

Solution: This is the memoryless property. Among standard discrete distributions, only the Geometric distribution is memoryless.

Concept Tested: Memoryless property (discrete analogue)

Question 6

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: $\text{Var}(X) = 5$. Find $\text{Var}(3X + 7)$.

Answer / Expected Answer: 45

Solution: $\text{Var}(aX+b) = a^2\text{Var}(X) = 9 \times 5 = 45$. The constant 7 doesn't affect variance.

Concept Tested: Variance of a linear transformation

Question 7

Type: MCQ Source: PYQ-PATTERN Difficulty: Medium

Question: A Binomial(n, p) distribution can be approximated by a Poisson distribution when:

Options: (A) n is small and p is close to 0.5 (B) n is large and p is large (C) n is large, p is small, and np remains moderate ($\approx \lambda$) (D) n and p are both close to 1

Answer: (C)

Solution: The Poisson approximation to Binomial holds under the "law of rare events": $n \rightarrow \infty, p \rightarrow 0$, with $np = \lambda$ fixed.

Concept Tested: Binomial-Poisson relationship

Question 8

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $f(x) = kx^2$ for $0 < x < 3$, and 0 otherwise, is a valid PDF. Find k .

Answer / Expected Answer: 0.111 (i.e., $1/9$)

Solution: $\int_0^3 kx^2 dx = k(27/3) = 9k = 1 \implies k = 1/9 \approx 0.111$.

Concept Tested: Validity condition of a PDF

Question 9

Type: MSQ Source: PYQ-PATTERN Difficulty: Medium

Question: Which of the following are true for any valid CDF $F(x)$ of a random variable X ?

Options: (A) F is non-decreasing (B) F is right-continuous (C) $\lim_{x \rightarrow -\infty} F(x) = 0$ (D) $\lim_{x \rightarrow \infty} F(x) = 1$

Answer: (A), (B), (C), (D)

Solution: All four are defining properties of a CDF for any random variable, discrete or continuous.

Concept Tested: Fundamental CDF properties

Question 10

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: The exponential distribution with rate λ is a special case of which distribution with shape parameter 1?

Options: (A) Beta (B) Gamma (C) Normal (D) Binomial

Answer: (B)

Solution: Gamma(shape=1, rate= λ) reduces exactly to Exponential(λ).

Concept Tested: Relationship between Gamma and Exponential

Question 11 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Easy

Question: $X \sim \text{Bernoulli}(p = 0.3)$. Find $\text{Var}(X)$.

Options: (A) 0.09 (B) 0.21 (C) 0.30 (D) 0.70

Answer: (B)

Solution: $\text{Var}(X) = p(1-p) = 0.3 \times 0.7 = 0.21$.

Concept Tested: Bernoulli variance

Question 12 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: $X \sim \text{Uniform}(0, 5)$. Find $P(X < 2)$.

Answer / Expected Answer: 0.4

Solution: For $U(0,5)$, $P(X < x) = x/5$. So $P(X < 2) = 2/5 = 0.4$.

Concept Tested: Uniform distribution CDF

Question 13 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: $X \sim \text{Binomial}(n=6, p=0.5)$. Find $P(X=3)$.

Options: (A) 0.2500 (B) 0.3125 (C) 0.4375 (D) 0.5000

Answer: (B)

Solution: $P(X=3) = C(6,3)(0.5)^3(0.5)^3 = 20/64 = 0.3125$.

Concept Tested: Binomial PMF calculation

Question 14 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $X \sim \text{Poisson}(\lambda=2)$. Find $P(X \geq 1)$ (round to 4 decimals).

Answer / Expected Answer: 0.8647

Solution: $P(X \geq 1) = 1 - P(X=0) = 1 - e^{-2} = 1 - 0.1353 = 0.8647$.

Concept Tested: Poisson complement rule

Question 15 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $f(x) = 2x$ for $0 < x < 1$ (0 otherwise). Find $E[X]$.

Answer / Expected Answer: 0.667 (i.e., $2/3$)

Solution: $E[X] = \int_0^1 x \cdot 2x \, dx = \int_0^1 2x^2 \, dx = 2/3 \approx 0.667$.

Concept Tested: Expectation of a continuous RV

Question 16 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: For $f(x) = 2x$, $0 < x < 1$, find $\text{Var}(X)$ (round to 4 decimals).

Answer / Expected Answer: 0.0556

Solution: $E[X^2] = \int_0^1 2x^3 \, dx = 0.5$. $E[X] = 2/3$. $\text{Var}(X) = 0.5 - (2/3)^2 = 0.5 - 0.4444 = 0.0556$.

Concept Tested: Variance via $E[X^2] - (E[X])^2$

Question 17 (Original)

Type: MSQ Source: ORIGINAL Difficulty: Medium

Question: For a Geometric distribution with $P(X=k) = (1-p)^{k-1}p$, $k = 1, 2, 3, \dots$, which of the following are TRUE?

Options: (A) Mean = $1/p$ (B) Variance = $(1-p)/p^2$ (C) It is memoryless (D) Its support starts at $k = 0$

Answer: (A), (B), (C)

Solution: The standard “number of trials until first success” version has support starting at $k=1$, not 0, so (D) is false. (A), (B), (C) are standard properties.

Concept Tested: Geometric distribution properties

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: $S = X_1 + X_2 + X_3 + X_4$ where X_i are i.i.d. $\text{Exponential}(\lambda=2)$. Find $\text{Var}(S)$.

Answer / Expected Answer: 1.0

Solution: $\text{Var}(X_i) = 1/\lambda^2 = 0.25$. For independent sum, $\text{Var}(S) = 4 \times 0.25 = 1$.

Concept Tested: Sum of i.i.d. exponentials (Gamma variance)

Question 19 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: $X \sim \text{Binomial}(n=5, p=0.6)$. Find $P(X \geq 4)$ (round to 4 decimals).

Answer / Expected Answer: 0.3370

Solution: $P(4) = C(5,4)(0.6)^4(0.4) = 5 \times 0.1296 \times 0.4 = 0.2592$. $P(5) = (0.6)^5 = 0.07776$. $P(X \geq 4) = 0.2592 + 0.07776 = 0.33696 \approx 0.3370$.

Concept Tested: Binomial tail probability, multi-step calculation

Question 20 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Hard

Question: Which of the following statements is FALSE?

Options: (A) For a continuous random variable X , $P(X=a) = 0$ for any real a (B) The CDF of a discrete random variable is always continuous (C) Variance of any random variable is always ≥ 0 (D) The derivative of the MGF at $t=0$ gives the mean

Answer: (B)

Solution: The CDF of a discrete random variable has jumps at each mass point, so it is a right-continuous step function, not continuous everywhere. All other statements are true.

Concept Tested: Conceptual traps — discrete vs. continuous CDF

Answer Key — Standard Distributions

1-C | 2-16/3 | 3-B | 4-0.0498 | 5-C | 6-45 | 7-C | 8-1/9 | 9-ABCD | 10-B | 11-B | 12-0.4 | 13-B | 14-0.8647 | 15-0.667 | 16-0.0556 | 17-ABC | 18-1.0 | 19-0.3370 | 20-B

2. Normal Distribution

Question 1

Type: NAT Source: PYQ (GATE DA 2024, memory-based) Difficulty: Easy

Question: A dataset has raw score $x = 75$, mean $\mu = 70$, standard deviation $\sigma = 5$. Compute the Z-score.

Answer / Expected Answer: 1.0

Solution: $z = (x - \mu) / \sigma = (75 - 70) / 5 = 1$.

Concept Tested: Z-score computation

Question 2

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: For the standard normal distribution, $P(Z < 1.96)$ is closest to:

Options: (A) 0.900 (B) 0.950 (C) 0.975 (D) 0.990

Answer: (C)

Solution: From the standard normal table, $\Phi(1.96) \approx 0.975$.

Concept Tested: Standard normal table lookup

Question 3

Type: NAT Source: PYQ-PATTERN (based on GATE CSE 2017-Set1 style) Difficulty: Hard

Question: $X \sim N(0, \sigma^2)$. Define $Y = \max(X, 0)$. Find the median of Y .

Answer / Expected Answer: 0

Solution: Since X is symmetric about 0, $P(X \leq 0) = 0.5$, so $P(Y=0) \geq 0.5$. The median of Y (smallest value m with $P(Y \leq m) \geq 0.5$) is therefore 0.

Concept Tested: Transformation of a normal RV, median under symmetry

Question 4

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: For a normal distribution, $P(\mu - 2\sigma < X < \mu + 2\sigma)$ is approximately:

Options: (A) 0.6826 (B) 0.9545 (C) 0.9973 (D) 1.0000

Answer: (B)

Solution: This is the empirical (68-95-99.7) rule: $\pm 2\sigma$ covers $\approx 95.45\%$.

Concept Tested: Empirical rule for normal distribution

Question 5

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim N(50, 100)$. Find $P(X > 60)$ (round to 4 decimals).

Answer / Expected Answer: 0.1587

Solution: $\sigma = 10$. $z = (60 - 50)/10 = 1$. $P(X > 60) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$.

Concept Tested: Standardization + tail probability

Question 6

Type: MSQ Source: PYQ-PATTERN Difficulty: Easy

Question: Which of the following are TRUE for a normal distribution?

Options: (A) It is symmetric about the mean (B) Mean = Median = Mode (C) Its excess kurtosis is 0 (D) Its skewness is 0

Answer: (A), (B), (C), (D)

Solution: All are standard shape properties of the normal distribution (kurtosis of a normal itself is 3; excess kurtosis, i.e., kurtosis - 3, is 0).

Concept Tested: Shape properties of normal distribution

Question 7

Type: MSQ Source: PYQ-PATTERN (based on GATE DA 2026 Q18 style) Difficulty: Medium

Question: Let $Z \sim N(0,1)$ with PDF $g(z)$ and CDF $G(z)$. Which of the following are TRUE?

Options: (A) $G'(z) = g(z)$ (B) $g(-z) = g(z)$ (C) $G(-z) = 1 - G(z)$ (D) $G(0) = 0$

Answer: (A), (B), (C)

Solution: G is the antiderivative of g (A true). g is symmetric/even (B true). By symmetry of standard normal, $G(-z) = 1 - G(z)$ (C true). $G(0) = 0.5$, not 0, so (D) is false.

Concept Tested: PDF-CDF relationship and symmetry of standard normal

Question 8

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim N(3, 4)$. $Y = 2X + 1$. Find $\text{Var}(Y)$.

Answer / Expected Answer: 16

Solution: $\text{Var}(Y) = 2^2 \cdot \text{Var}(X) = 4 \times 4 = 16$.

Concept Tested: Linear transformation of a normal RV

Question 9

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: $X \sim N(2, 3)$ and $Y \sim N(5, 4)$ are independent. Let $Z = X + Y$. Find the mean of Z .

Options: (A) 3 (B) 5 (C) 7 (D) 10

Answer: (C)

Solution: For independent normals, means add: $E[Z] = 2 + 5 = 7$.

Concept Tested: Sum of independent normal random variables

Question 10

Type: NAT Source: PYQ-PATTERN (based on GATE DA 2026 Q34 style) Difficulty: Medium

Question: $X \sim \text{Bernoulli}(p=0.3)$ and $Y \sim N(0, 4)$ are independent. Let $W = X + Y$. Find $\text{Var}(W)$.

Answer / Expected Answer: 4.21

Solution: $\text{Var}(X) = p(1-p) = 0.21$. $\text{Var}(Y) = 4$. Independence $\implies \text{Var}(W) = 0.21 + 4 = 4.21$.

Concept Tested: Variance of sum of independent RVs (mixed types)

Question 11 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: $X \sim N(10, 4)$. Find the z-score for $x = 14$.

Answer / Expected Answer: 2.0

Solution: $\sigma = 2$. $z = (14 - 10)/2 = 2$.

Concept Tested: Z-score computation

Question 12 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Easy

Question: The standard normal distribution has parameters:

Options: (A) $\mu=1, \sigma^2=0$ (B) $\mu=0, \sigma^2=1$ (C) $\mu=0, \sigma^2=0$ (D) $\mu=1, \sigma^2=1$

Answer: (B)

Solution: By definition, standard normal is $N(0, 1)$.

Concept Tested: Definition of standard normal

Question 13 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $X \sim N(100, 225)$. Find $P(X < 115)$ (round to 4 decimals).

Answer / Expected Answer: 0.8413

Solution: $\sigma=15$. $z=(115-100)/15=1$. $P(X < 115) = \Phi(1) = 0.8413$.

Concept Tested: Standardization

Question 14 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: For $Z \sim N(0, 1)$, $P(-1 < Z < 1)$ is closest to:

Options: (A) 0.5000 (B) 0.6826 (C) 0.8413 (D) 0.9545

Answer: (B)

Solution: This is the $\pm 1\sigma$ empirical-rule value: $\approx 68.26\%$.

Concept Tested: Interval probability for standard normal

Question 15 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $X \sim N(50, 100)$. Find x such that $P(X < x) = 0.90$ (round to 2 decimals).

Answer / Expected Answer: 62.82

Solution: $z_{0.90} \approx 1.2816$. $x = 50 + 1.2816 \times 10 = 62.816 \approx 62.82$.

Concept Tested: Inverse normal (percentile) calculation

Question 16 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: $X \sim N(4, 9)$ and $Y \sim N(4, 9)$ are independent and identically distributed. Find $P(X > Y)$.

Options: (A) 0 (B) 0.25 (C) 0.50 (D) 0.75

Answer: (C)

Solution: $X - Y$ is normal with mean 0 and is symmetric about 0, so $P(X > Y) = P(X - Y > 0) = 0.5$.

Concept Tested: Symmetry argument for i.i.d. normals

Question 17 (Original)

Type: MSQ Source: ORIGINAL Difficulty: Medium

Question: Regarding the z-score of an observation, which of the following are TRUE?

Options: (A) The standardized variable has mean 0 and variance 1 (B) A z-score is unit-less (C) A negative z-score means the value is below the mean (D) Standardizing changes the shape of the original distribution

Answer: (A), (B), (C)

Solution: Standardization is a linear shift-and-scale operation; it does not change the shape (skewness/kurtosis) of the distribution, so (D) is false.

Concept Tested: Meaning and properties of standardization

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: $X \sim N(0, 1)$. Find $P(|X| > 1.5)$ (round to 4 decimals).

Answer / Expected Answer: 0.1336

Solution: $P(|X| > 1.5) = 2(1 - \Phi(1.5)) = 2(1 - 0.9332) = 2 \times 0.0668 = 0.1336$.

Concept Tested: Two-tailed probability calculation

Question 19 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Hard

Question: For $X \sim N(\mu, \sigma^2)$, which is TRUE?

Options: (A) $P(X > \mu + \sigma) > P(X < \mu - 2\sigma)$ (B) $P(X > \mu + \sigma) < P(X < \mu - 2\sigma)$ (C) $P(X > \mu + \sigma) = P(X < \mu - 2\sigma)$ (D) Cannot be determined without knowing σ

Answer: (A)

Solution: $P(X > \mu + \sigma) = 1 - \Phi(1) = 0.1587$. $P(X < \mu - 2\sigma) = 1 - \Phi(2) = 0.0228$. Since $0.1587 > 0.0228$, (A) holds regardless of σ (it cancels out in standardization).

Concept Tested: Comparing tail probabilities

Question 20 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: $X \sim N(20, 25)$. Find a such that $P(X < a) = 0.20$ (round to 2 decimals).

Answer / Expected Answer: 15.79

Solution: $z_{0.20} \approx -0.8416$. $a = 20 + (-0.8416)(5) = 20 - 4.208 = 15.792 \approx 15.79$.

Concept Tested: Inverse normal for a lower-tail percentile

Answer Key — Normal Distribution

1-1.0 | 2-C | 3-0 | 4-B | 5-0.1587 | 6-ABCD | 7-ABC | 8-16 | 9-C | 10-4.21 | 11-2.0 | 12-B | 13-0.8413 | 14-B | 15-62.82 | 16-C | 17-ABC | 18-0.1336 | 19-A | 20-15.79

3. Exponential Distribution

Question 1

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: $X \sim \text{Exponential}(\lambda=2)$. Find $E[X]$.

Answer / Expected Answer: 0.5

Solution: Mean of $\text{Exponential}(\lambda) = 1/\lambda = 1/2 = 0.5$.

Concept Tested: Exponential mean

Question 2

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: $X \sim \text{Exponential}(\lambda=2)$. Find $\text{Var}(X)$.

Answer / Expected Answer: 0.25

Solution: $\text{Var}(X) = 1/\lambda^2 = 1/4 = 0.25$.

Concept Tested: Exponential variance

Question 3

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: Which continuous distribution satisfies $P(X > s+t | X > s) = P(X > t)$?

Options: (A) Normal (B) Uniform (C) Exponential (D) Gamma (shape > 1)

Answer: (C)

Solution: This memoryless property uniquely characterizes the Exponential distribution among continuous distributions.

Concept Tested: Memoryless property

Question 4

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim \text{Exponential}(\lambda=0.5)$. Find $P(X > 2)$ (round to 4 decimals).

Answer / Expected Answer: 0.3679

Solution: $P(X > x) = e^{-\lambda x} = e^{-1} \approx 0.3679$.

Concept Tested: Exponential tail probability

Question 5

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: The CDF of $\text{Exponential}(\lambda)$ for $x \geq 0$ is:

Options: (A) $F(x) = \lambda e^{-\lambda x}$ (B) $F(x) = 1 - e^{-\lambda x}$ (C) $F(x) = e^{-\lambda x}$ (D) $F(x) = 1 - \lambda e^{-\lambda x}$

Answer: (B)

Solution: Integrating the PDF $\lambda e^{-\lambda x}$ from 0 to x gives $F(x) = 1 - e^{-\lambda x}$.

Concept Tested: Exponential CDF derivation

Question 6

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim \text{Exponential}$ with mean 5. Find $P(X < 10)$ (round to 4 decimals).

Answer / Expected Answer: 0.8647

Solution: $\lambda = 1/5 = 0.2$. $P(X < 10) = 1 - e^{-(0.2 \times 10)} = 1 - e^{-2} = 1 - 0.1353 = 0.8647$.

Concept Tested: Converting mean to rate parameter

Question 7

Type: MSQ Source: PYQ-PATTERN Difficulty: Medium

Question: Which of the following are TRUE about the Exponential distribution?

Options: (A) It is memoryless (B) It is the only continuous distribution with the memoryless property (C) Its mean always equals its variance (D) Its mode is 0

Answer: (A), (B), (D)

Solution: Mean = $1/\lambda$, Variance = $1/\lambda^2$; these are equal only when $\lambda=1$, so (C) is false in general. (A), (B), (D) are standard properties.

Concept Tested: Distinguishing true vs. commonly-confused properties

Question 8

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim \text{Exponential}(\lambda_1=2)$ and $Y \sim \text{Exponential}(\lambda_2=3)$, independent. Find $E[\min(X,Y)]$.

Answer / Expected Answer: 0.2

Solution: $\min(X,Y) \sim \text{Exponential}(\lambda_1+\lambda_2) = \text{Exponential}(5)$. Mean = $1/5 = 0.2$.

Concept Tested: Minimum of independent exponentials

Question 9

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: In a Poisson process with rate λ , the time between successive arrivals follows:

Options: (A) Normal(0, λ) (B) Poisson(λ) (C) Exponential(λ) (D) Uniform(0, λ)

Answer: (C)

Solution: Inter-arrival times of a Poisson process are i.i.d. Exponential(λ).

Concept Tested: Poisson process – Exponential connection

Question 10

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $X \sim \text{Exponential}(\lambda=4)$. Find the median of X (round to 4 decimals).

Answer / Expected Answer: 0.1733

Solution: Median = $\ln(2)/\lambda = 0.6931/4 \approx 0.1733$.

Concept Tested: Median of exponential distribution

Question 11 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: $X \sim \text{Exponential}(\lambda=1)$. Find $P(X < 1)$ (round to 4 decimals).

Answer / Expected Answer: 0.6321

Solution: $P(X < 1) = 1 - e^{-1} \approx 0.6321$.

Concept Tested: Basic CDF evaluation

Question 12 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Easy

Question: As λ increases for an Exponential(λ) distribution, the mean:

Options: (A) Increases (B) Decreases (C) Stays the same (D) Becomes negative

Answer: (B)

Solution: Mean = $1/\lambda$, which decreases as λ increases.

Concept Tested: Parameter-mean relationship

Question 13 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $X \sim \text{Exponential}$ with mean 4. Find $P(2 < X < 6)$ (round to 4 decimals).

Answer / Expected Answer: 0.3834

Solution: $\lambda=0.25$. $P(2 < X < 6) = e^{-(0.25 \cdot 2)} - e^{-(0.25 \cdot 6)} = 0.6065 - 0.2231 = 0.3834$.

Concept Tested: Interval probability calculation

Question 14 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: For $X \sim \text{Exponential}(\lambda)$, it is known that $P(X > 5) = 0.2$. Using the memoryless property, find $P(X > 8 \mid X > 3)$.

Answer / Expected Answer: 0.2

Solution: By memorylessness, $P(X > 8 \mid X > 3) = P(X > 5) = 0.2$ (since $8 - 3 = 5$).

Concept Tested: Applying memoryless property numerically

Question 15 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $X \sim \text{Exponential}(\lambda=2)$. Find $P(X > 1 \mid X > 0.5)$ (round to 4 decimals).

Answer / Expected Answer: 0.3679

Solution: By memorylessness, $P(X > 1 \mid X > 0.5) = P(X > 0.5) = e^{-(2 \times 0.5)} = e^{-1} \approx 0.3679$.

Concept Tested: Memoryless property applied to shifted intervals

Question 16 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: For $X \sim \text{Exponential}(\lambda)$, which expression correctly gives $E[X^2]$?

Options: (A) $1/\lambda^2$ (B) $2/\lambda^2$ (C) $1/\lambda$ (D) $2/\lambda$

Answer: (B)

Solution: $\text{Var}(X) = 1/\lambda^2$ and $(E[X])^2 = 1/\lambda^2$, so $E[X^2] = \text{Var}(X) + (E[X])^2 = 1/\lambda^2 + 1/\lambda^2 = 2/\lambda^2$.

Concept Tested: Second moment from variance and mean

Question 17 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: For $X \sim \text{Exponential}(\lambda)$, $P(X > 10) = 0.1$. Find λ (round to 4 decimals).

Answer / Expected Answer: 0.2303

Solution: $e^{-(10\lambda)} = 0.1 \implies -10\lambda = \ln(0.1) \implies \lambda = -\ln(0.1)/10 = 2.3026/10 = 0.2303$.

Concept Tested: Solving for the rate parameter

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: X, Y are i.i.d. Exponential($\lambda=1$) and independent. Find $P(X < Y)$.

Answer / Expected Answer: 0.5

Solution: By symmetry between two i.i.d. continuous variables, $P(X < Y) = P(Y < X) = 0.5$ ($P(X=Y)=0$).

Concept Tested: Symmetry argument for i.i.d. continuous RVs

Question 19 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Hard

Question: The time between calls at a helpdesk is Exponential with mean 10 minutes. Given that no call has arrived in the last 5 minutes, the probability of waiting more than an additional 10 minutes is:

Options: (A) e^{-1} (B) $e^{-1.5}$ (C) $e^{-0.5}$ (D) $1 - e^{-1}$

Answer: (A)

Solution: By memorylessness, the extra wait beyond the already-elapsed 5 minutes still behaves like a fresh Exponential(mean=10). So the answer is $P(X > 10) = e^{-(10/10)} = e^{-1}$.

Concept Tested: Applying memorylessness in a word problem

Question 20 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: A component's lifetime is Exponential($\lambda=0.02$ per hour). Given it has survived 50 hours, find the probability it survives another 100 hours (round to 4 decimals).

Answer / Expected Answer: 0.1353

Solution: By memorylessness, this equals the unconditional probability of surviving 100 hours: $P(X > 100) = e^{-(0.02 \times 100)} = e^{-2} \approx 0.1353$.

Concept Tested: Reliability/survival application of memorylessness

Answer Key — Exponential Distribution

1-0.5 | 2-0.25 | 3-C | 4-0.3679 | 5-B | 6-0.8647 | 7-ABD | 8-0.2 | 9-C | 10-0.1733 | 11-0.6321 | 12-B | 13-0.3834 | 14-0.2 | 15-0.3679 | 16-B | 17-0.2303 | 18-0.5 | 19-A | 20-0.1353

4. Joint Distribution

Question 1

Type: MCQ Source: PYQ (based on GATE DS&AI 2024, Q14) Difficulty: Medium

Question: Random variables U, V, W, X, Y have joint distribution satisfying: $P(U, V, W, X, Y) = P(U) \cdot P(V) \cdot P(W|U, V) \cdot P(X|W) \cdot P(Y|W)$ Which conditional independence relation is implied by this factorization?

Options: (A) X and Y are independent (unconditionally) (B) $X \perp Y \mid W$ (C) U and V are independent given W (D) $X \perp Y \mid U, V$

Answer: (B)

Solution: Once W is known, X depends only on W and Y depends only on W in the factorization, so X and Y become conditionally independent given W: $X \perp Y \mid W$.

Concept Tested: Reading conditional independence from a factorized joint distribution

Question 2

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: $f(x, y) = x + y$ for $0 < x < 1$, $0 < y < 1$ (0 otherwise). Find the marginal $f_X(x)$ at $x = 0.4$.

Answer / Expected Answer: 0.9

Solution: $f_X(x) = \int_0^1 (x + y) dy = x + 0.5$. At $x=0.4$: $0.4 + 0.5 = 0.9$.

Concept Tested: Marginal density from joint PDF

Question 3

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Using $f(x,y) = x+y$ on the unit square (Q2 above), find $E[X]$ (round to 4 decimals).

Answer / Expected Answer: 0.5833

Solution: $\int_0^1 \int_0^{1-x} x(x+y) dy dx = \int_0^1 x(x+0.5) dx = \int_0^1 (x^2+0.5x) dx = 1/3 + 0.25 = 0.5833$.

Concept Tested: Expectation using marginal PDF

Question 4

Type: NAT Source: PYQ-PATTERN Difficulty: Hard

Question: Using $f(x,y) = x+y$ on the unit square, find $P(X<0.5, Y<0.5)$.

Answer / Expected Answer: 0.09375

Solution: $\int_0^{0.5} \int_0^{0.5} (x+y) dy dx$: inner integral = $0.5x+0.0625$. Outer: $\int_0^{0.5} (0.5x+0.0625) dx = 0.0625+0.03125 = 0.09375$.

Concept Tested: Joint probability over a sub-region

Question 5

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Joint PMF table:

X	0	1
0	0.1	0.2
1	0.3	0.4

Find $\text{Cov}(X,Y)$ (round to 4 decimals).

Answer / Expected Answer: -0.02

Solution: $P(X=1)=0.7$, $P(Y=1)=0.6$, $E[XY]=P(X=1,Y=1)=0.4$. $\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = 0.4 - (0.7 \times 0.6) = 0.4 - 0.42 = -0.02$.

Concept Tested: Covariance from a joint PMF table

Question 6

Type: MCQ Source: PYQ-PATTERN Difficulty: Medium

Question: Using the table in Q5, are X and Y independent?

Options: (A) Yes, since $P(X=0,Y=0)=P(X=0)P(Y=0)$ (B) No, since $P(X=0,Y=0)=0.1$ but $P(X=0)P(Y=0)=0.12$ (C) Yes, since $\text{Cov}(X,Y) \neq 0$ (D) Cannot be determined

Answer: (B)

Solution: $P(X=0)=0.3$, $P(Y=0)=0.4$, product= $0.12 \neq 0.1 = P(X=0,Y=0)$, so they are not independent.

Concept Tested: Checking independence from a joint table

Question 7

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: Using the table in Q5, find the marginal $P(X=0)$.

Answer / Expected Answer: 0.3

Solution: $P(X=0) = 0.1+0.2 = 0.3$.

Concept Tested: Marginal probability from a joint table

Question 8

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Using the table in Q5, find $P(Y=1 | X=1)$ (round to 4 decimals).

Answer / Expected Answer: 0.5714

Solution: $P(Y=1|X=1) = P(X=1,Y=1)/P(X=1) = 0.4/0.7 = 0.5714$.

Question 9

Type: MSQ Source: PYQ-PATTERN Difficulty: Medium

Question: Which of the following statements about independence and covariance are TRUE in general?

Options: (A) If X, Y are independent, then $\text{Cov}(X,Y)=0$ (B) If $\text{Cov}(X,Y)=0$, then X, Y are independent (C) If X, Y are independent, then $E[XY]=E[X]E[Y]$ (D) Independence always implies zero correlation coefficient

Answer: (A), (C), (D)

Solution: Independence $\implies$ zero covariance/correlation and $E[XY]=E[X]E[Y]$. The converse (B) is false in general — zero covariance only rules out linear dependence, not all dependence.

Concept Tested: Independence vs. uncorrelatedness — conceptual trap

Question 10

Type: MCQ Source: PYQ-PATTERN (based on GATE DS&AI 2024 factorization theme) Difficulty: Hard

Question: Given $P(U,V,W,X,Y) = P(U)P(V)P(W|U,V)P(X|W)P(Y|W)$, which pair is generally NOT independent (even unconditionally) due to the shared parent W ?

Options: (A) U and V (B) X and Y (C) U and Y (D) V and X

Answer: (B)

Solution: X and Y are unconditionally dependent because they share the common cause W (a “collider”-free common-parent structure induces marginal dependence); they only become independent once conditioned on W .

Concept Tested: Marginal vs. conditional independence in graphical/factorized models

Question 11 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: Joint PMF table:

X	1	2
1	0.2	0.1
2	0.3	0.4

Find $E[X]$.

Answer / Expected Answer: 1.7

Solution: $P(X=1)=0.3, P(X=2)=0.7. E[X] = 1(0.3)+2(0.7) = 1.7.$

Concept Tested: Expectation from a joint table’s marginal

Question 12 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: Using the table in Q11, find $P(Y=2)$.

Answer / Expected Answer: 0.5

Solution: $P(Y=2) = 0.1+0.4 = 0.5.$

Concept Tested: Marginal probability

Question 13 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Using the table in Q11, find $E[XY]$.

Answer / Expected Answer: 2.6

Solution: $E[XY] = (1)(1)(0.2)+(1)(2)(0.1)+(2)(1)(0.3)+(2)(2)(0.4) = 0.2+0.2+0.6+1.6 = 2.6.$

Concept Tested: Expectation of a product from joint PMF

Question 14 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Using the table in Q11, find $\text{Cov}(X,Y)$.

Answer / Expected Answer: 0.05

Solution: $E[Y] = 1(0.5) + 2(0.5) = 1.5$. $\text{Cov}(X,Y) = E[XY] - E[X]E[Y] = 2.6 - (1.7 \times 1.5) = 2.6 - 2.55 = 0.05$.

Concept Tested: Covariance calculation

Question 15 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: Are X and Y in the table of Q11 independent?

Options: (A) Yes (B) No (C) Only if $\text{Cov}(X,Y)=0$ (D) Cannot say without more data

Answer: (B)

Solution: $P(X=1,Y=1)=0.2$ but $P(X=1)P(Y=1)=0.3 \times 0.5=0.15 \neq 0.2$, so they are not independent.

Concept Tested: Independence check via joint vs. product of marginals

Question 16 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: $f(x,y) = c$ for $0 < x < 2$, $0 < y < 1$ (a uniform density on a rectangle). Find c.

Answer / Expected Answer: 0.5

Solution: Total area = $2 \times 1 = 2$. For a valid PDF, $c \times 2 = 1 \implies c = 0.5$.

Concept Tested: Normalizing constant for uniform joint density

Question 17 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Using $f(x,y)=0.5$ from Q16, find $P(X < 1, Y < 0.5)$.

Answer / Expected Answer: 0.25

Solution: Region area = $1 \times 0.5 = 0.5$. Probability = density $\times$ area = $0.5 \times 0.5 = 0.25$.

Concept Tested: Probability over a sub-rectangle of a uniform joint density

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: $f(x,y) = 8xy$ for $0 < x < y < 1$ (0 otherwise). Find $E[X]$ (round to 4 decimals).

Answer / Expected Answer: 0.5333

Solution: $f_X(x) = \int_x^1 8xy \, dy = 8x[(1-x^2)/2] = 4x-4x^3$. $E[X] = \int_0^1 x(4x-4x^3) \, dx = \int_0^1 (4x^2-4x^4) \, dx = 4/3 - 4/5 = 8/15 \approx 0.5333$.

Concept Tested: Marginal and expectation over a triangular support region

Question 19 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: Using $f(x,y) = 8xy$ on $0 < x < y < 1$ (Q18), find $P(X < 0.5)$.

Answer / Expected Answer: 0.4375

Solution: $f_X(x) = 4x-4x^3$. $P(X < 0.5) = \int_0^{0.5} (4x-4x^3) \, dx = [2x^2-x^4]_0^{0.5} = 0.5 - 0.0625 = 0.4375$.

Concept Tested: Marginal CDF over a non-rectangular support

Question 20 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: X, Y are jointly normal with $\text{Var}(X)=4$, $\text{Var}(Y)=9$, correlation $\rho=0.6$. Find $\text{Var}(X+Y)$.

Answer / Expected Answer: 20.2

Solution: $\text{Cov}(X,Y) = \rho \cdot \sigma_X \cdot \sigma_Y = 0.6 \times 2 \times 3 = 3.6$. $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X,Y) = 4 + 9 + 7.2 = 20.2$.

Concept Tested: Variance of a sum with non-zero covariance

Answer Key — Joint Distribution

1-B | 2-0.9 | 3-0.5833 | 4-0.09375 | 5-(-0.02) | 6-B | 7-0.3 | 8-0.5714 | 9-ACD | 10-B | 11-1.7 | 12-0.5 | 13-2.6 | 14-0.05 | 15-B | 16-0.5 | 17-0.25 | 18-0.5333 | 19-0.4375 | 20-20.2

5. Naive Bayes Classifier

Question 1

Type: MCQ Source: PYQ (based on GATE DS&AI 2024, Q10) Difficulty: Medium

Question: A dataset has K binary-valued attributes ($K > 2$) for a two-class classification task. How many parameters must be estimated for a Naive Bayes classifier (including the class prior)?

Options: (A) $K+1$ (B) $2K$ (C) $2K+1$ (D) K^2+1

Answer: (C)

Solution: Each class needs K Bernoulli parameters (one per attribute) $\rightarrow 2K$ parameters for both classes. The class prior needs 1 more parameter ($P(Y=1)$, since $P(Y=0) = 1 - P(Y=1)$). Total = $2K+1$.

Concept Tested: Parameter counting in Naive Bayes

Question 2

Type: NAT Source: PYQ-PATTERN (based on GATE DA 2025, Q25 style) Difficulty: Medium

Question: Two-class Naive Bayes with $P(y_1)=0.6$, $P(y_2)=0.4$. Given $P(x_1=1|y_1)=0.7$, $P(x_2=1|y_1)=0.5$, $P(x_1=1|y_2)=0.3$, $P(x_2=1|y_2)=0.8$. For test point $(x_1=1, x_2=1)$, find the posterior $P(y_1 | x)$ (round to 4 decimals).

Answer / Expected Answer: 0.6863

Solution: $\text{Score}(y_1) = 0.6 \times 0.7 \times 0.5 = 0.21$. $\text{Score}(y_2) = 0.4 \times 0.3 \times 0.8 = 0.096$. $P(y_1|x) = 0.21 / (0.21 + 0.096) = 0.21 / 0.306 \approx 0.6863$.

Concept Tested: Full posterior computation with normalization

Question 3

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: In a class with $N=10$ training examples, a feature value was seen 3 times. Using Laplace (add-one) smoothing with vocabulary size $V=5$, find the smoothed probability.

Answer / Expected Answer: 0.2667

Solution: $P_{\text{smoothed}} = (\text{count}+1)/(N+V) = (3+1)/(10+5) = 4/15 \approx 0.2667$.

Concept Tested: Laplace smoothing formula

Question 4

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: Without smoothing, what happens if a feature value was never observed for a given class during training?

Options: (A) The likelihood for that feature is set to 1 automatically (B) The posterior for that class becomes 0 regardless of other evidence (C) The classifier ignores that feature (D) The prior is recalculated to compensate

Answer: (B)

Solution: A zero likelihood term makes the entire product (posterior score) zero for that class, no matter how strong other features are — the “zero-frequency problem.”

Concept Tested: Zero-frequency problem

Question 5

Type: MSQ Source: PYQ-PATTERN Difficulty: Easy

Question: Which of the following are TRUE about the Naive Bayes classifier?

Options: (A) It assumes features are conditionally independent given the class (B) It is a generative model (C) It always requires very large training datasets to work well (D) It directly models $P(Y|X)$ without any assumption about $P(X|Y)$

Answer: (A), (B)

Solution: Naive Bayes is generative (models $P(X|Y)P(Y)$) and assumes conditional independence. It is known to work reasonably even with small datasets, so (C) is false; (D) contradicts the generative nature of the model.

Concept Tested: Fundamental characterization of Naive Bayes

Question 6

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Using $\text{Score}(y_1)=0.21$ from Q2, find $\log_{10}(0.21)$ (round to 4 decimals).

Answer / Expected Answer: -0.6778

Solution: $\log_{10}(0.21) = \log_{10}(21) - 2 = 1.3222 - 2 = -0.6778$.

Concept Tested: Log-probability calculation

Question 7

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: In Gaussian Naive Bayes, each continuous feature is assumed, within each class, to follow:

Options: (A) Uniform distribution (B) Exponential distribution (C) Normal distribution (D) Poisson distribution

Answer: (C)

Solution: Gaussian Naive Bayes models each feature as $\text{Normal}(\mu_{\text{class}}, \sigma^2_{\text{class}})$ conditional on the class.

Concept Tested: Gaussian Naive Bayes assumption

Question 8

Type: NAT Source: PYQ-PATTERN (based on GATE DS&AI 2024 parameter-counting theme) Difficulty: Hard

Question: A Naive Bayes classifier has $K=3$ categorical attributes, each taking $m=4$ possible values, for a two-class problem. How many parameters must be estimated in total (including the prior)?

Answer / Expected Answer: 19

Solution: Each attribute needs $(m-1)=3$ free parameters per class. For $K=3$ attributes and 2 classes: $2 \times K \times (m-1) = 2 \times 3 \times 3 = 18$, plus 1 for the class prior = 19.

Concept Tested: Parameter counting for multi-valued categorical attributes

Question 9

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: Given $P(B|A)=0.8$, $P(A)=0.3$, $P(B)=0.5$, find $P(A|B)$.

Answer / Expected Answer: 0.48

Solution: By Bayes' theorem: $P(A|B) = P(B|A)P(A)/P(B) = (0.8 \times 0.3)/0.5 = 0.24/0.5 = 0.48$.

Concept Tested: Basic Bayes' theorem

Question 10

Type: MCQ Source: PYQ-PATTERN Difficulty: Medium

Question: In Bayes' theorem $P(Y|X) = P(X|Y)P(Y)/P(X)$, the "evidence" term $P(X)$ is:

Options: (A) The most important term for determining which class wins (B) A normalizing constant, and can be dropped when only comparing classes for the same X (C) Always equal to 1 (D) Different for each class being compared

Answer: (B)

Solution: $P(X)$ is identical across all classes for a fixed input, so it doesn't affect which class has the highest posterior — it's only needed if the actual probability value (not just the argmax) is required.

Concept Tested: Role of the evidence term in classification

Question 11 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: $P(\text{spam})=0.4$, $P(\text{"word"}|\text{spam})=0.6$. Find the unnormalized Naive Bayes score for the spam class given this one word is present.

Answer / Expected Answer: 0.24

Solution: $\text{Score}(\text{spam}) = P(\text{spam}) \times P(\text{word}|\text{spam}) = 0.4 \times 0.6 = 0.24$.

Concept Tested: Unnormalized class score computation

Question 12 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Easy

Question: $P(\text{not spam})=0.6$, $P(\text{"word"}|\text{not spam})=0.2$. Using Q11's score of 0.24 for spam, which class does Naive Bayes predict?

Options: (A) Spam (B) Not spam (C) Tie (D) Cannot determine

Answer: (A)

Solution: $\text{Score}(\text{not spam}) = 0.6 \times 0.2 = 0.12$. Since $0.24 > 0.12$, spam is predicted.

Concept Tested: Comparing unnormalized class scores

Question 13 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Using scores 0.24 (spam) and 0.12 (not spam) from Q11-Q12, find the normalized posterior $P(\text{spam}|\text{word})$ (round to 4 decimals).

Answer / Expected Answer: 0.6667

Solution: $P(\text{spam}|\text{word}) = 0.24 / (0.24 + 0.12) = 0.24 / 0.36 \approx 0.6667$.

Concept Tested: Posterior normalization

Question 14 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Two classes, three independent binary features. $P(y_1)=0.5$, $P(x_1=1|y_1)=0.9$, $P(x_2=1|y_1)=0.8$, $P(x_3=1|y_1)=0.3$. Test point: $x_1=1$, $x_2=1$, $x_3=0$. Find the Naive Bayes score for y_1 .

Answer / Expected Answer: 0.252

Solution: $\text{Score}(y_1) = P(y_1) \cdot P(x_1=1|y_1) \cdot P(x_2=1|y_1) \cdot P(x_3=0|y_1) = 0.5 \times 0.9 \times 0.8 \times 0.7 = 0.252$.

Concept Tested: Multi-feature Naive Bayes score

Question 15 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Same setup as Q14: $P(y_2)=0.5$, $P(x_1=1|y_2)=0.4$, $P(x_2=1|y_2)=0.3$, $P(x_3=1|y_2)=0.4$. Find the Naive Bayes score for y_2 on the same test point ($x_1=1, x_2=1, x_3=0$).

Answer / Expected Answer: 0.036

Solution: $\text{Score}(y_2) = 0.5 \times 0.4 \times 0.3 \times 0.6 = 0.036$.

Concept Tested: Multi-feature Naive Bayes score, comparison setup

Question 16 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Using scores 0.252 (y_1) and 0.036 (y_2) from Q14-Q15, find the posterior $P(y_1|x)$ (round to 4 decimals).

Answer / Expected Answer: 0.875

Solution: $P(y_1|x) = 0.252 / (0.252 + 0.036) = 0.252 / 0.288 = 0.875$.

Concept Tested: Normalized posterior in a multi-feature setting

Question 17 (Original)

Type: MSQ Source: ORIGINAL Difficulty: Medium

Question: If the conditional-independence assumption of Naive Bayes is dropped (a full joint model is used instead), which of the following are TRUE?

Options: (A) The number of parameters to estimate grows exponentially with the number of features (B) The model may capture feature correlations that Naive Bayes misses (C) Accuracy is guaranteed to improve on every dataset (D) Computational and data requirements typically increase

Answer: (A), (B), (D)

Solution: Modeling the full joint $P(X|Y)$ requires exponentially many parameters and more data/computation, and can capture correlations — but there's no guarantee of better accuracy on every dataset (bias-variance tradeoff), so (C) is false.

Concept Tested: Consequences of the independence assumption

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: A feature value has never appeared in a class with 20 training examples. Using Laplace smoothing with vocabulary size 4, find the smoothed probability.

Answer / Expected Answer: 0.0417

Solution: $P_{\text{smoothed}} = (0+1)/(20+4) = 1/24 \approx 0.0417$.

Concept Tested: Laplace smoothing under the zero-frequency scenario

Question 19 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: Class C1 has 15 samples, with a word appearing in 6 of them. Class C2 has 25 samples, with the word appearing in 2. Priors: $P(C1)=15/40$, $P(C2)=25/40$. Using Laplace smoothing (vocabulary size=2, for presence/absence), find the posterior $P(C1 \mid \text{word present})$ (round to 4 decimals).

Answer / Expected Answer: 0.6899

Solution: $P(\text{word}|C1) = (6+1)/(15+2) = 7/17 \approx 0.4118$. $P(\text{word}|C2) = (2+1)/(25+2) = 3/27 \approx 0.1111$. $\text{Score}(C1) = 0.375 \times 0.4118 = 0.1544$. $\text{Score}(C2) = 0.625 \times 0.1111 = 0.0694$. $P(C1|\text{word}) = 0.1544/(0.1544+0.0694) \approx 0.6899$.

Concept Tested: End-to-end Laplace-smoothed Naive Bayes classification

Question 20 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Hard

Question: Naive Bayes implementations typically compute $\log P(y) + \sum \log P(x_i|y)$ instead of the raw product. The main reason is:

Options: (A) Logarithms make the independence assumption valid (B) It avoids numerical underflow from multiplying many small probabilities (C) It changes which class is predicted, making results more accurate (D) It removes the need for a prior

Answer: (B)

Solution: Multiplying many probabilities (each <1) can underflow to 0 in floating-point arithmetic. Since log is monotonic, argmax is preserved, but underflow is avoided by summing logs instead.

Concept Tested: Numerical stability rationale for log-space computation

Answer Key — Naive Bayes Classifier

1-C | 2-0.6863 | 3-0.2667 | 4-B | 5-AB | 6-(-0.6778) | 7-C | 8-19 | 9-0.48 | 10-B | 11-0.24 | 12-A | 13-0.6667 | 14-0.252 | 15-0.036 | 16-0.875 | 17-ABD | 18-0.0417 | 19-0.6899 | 20-B

6. Number Systems

Question 1

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the remainder when 7^{100} is divided by 5.

Answer / Expected Answer: 1

Solution: $7 \equiv 2 \pmod{5}$. Cycle of 2 mod 5: 2,4,3,1 (period 4). $100 \bmod 4 = 0 \rightarrow$ use the 4th position $\rightarrow$ remainder 1.

Concept Tested: Cyclicity / modular arithmetic

Question 2

Type: NAT Source: PYQ-PATTERN Difficulty: Easy

Question: Find the number of factors of 360.

Answer / Expected Answer: 24

Solution: $360 = 2^3 \times 3^2 \times 5^1$. Number of factors = $(3+1)(2+1)(1+1) = 4 \times 3 \times 2 = 24$.

Concept Tested: Number of factors formula

Question 3

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the sum of all factors of 28.

Answer / Expected Answer: 56

Solution: $28 = 2^2 \times 7$. Sum of factors = $(1+2+4)(1+7) = 7 \times 8 = 56$.

Concept Tested: Sum of factors formula

Question 4

Type: MCQ Source: PYQ-PATTERN Difficulty: Easy

Question: Find LCM(18, 24).

Options: (A) 36 (B) 54 (C) 72 (D) 144

Answer: (C)

Solution: $\text{HCF}(18,24)=6$. $\text{LCM} = (18 \times 24)/6 = 432/6 = 72$.

Concept Tested: HCF-LCM relationship

Question 5

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the last digit of 7^{123} .

Answer / Expected Answer: 3

Solution: Cycle of last digit of 7: 7,9,3,1 (period 4). $123 \bmod 4 = 3 \rightarrow$ 3rd position $\rightarrow$ last digit 3.

Concept Tested: Cyclicity of last digits

Question 6

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the remainder when 2^{50} is divided by 7.

Answer / Expected Answer: 4

Solution: Cycle of 2 mod 7: 2,4,1 (period 3). $50 \bmod 3 = 2 \rightarrow$ 2nd position $\rightarrow$ remainder 4.

Concept Tested: Modular exponentiation via cyclicity

Question 7

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the number of trailing zeros in $100!$.

Answer / Expected Answer: 24

Solution: Count factors of 5: $[100/5] + [100/25] = 20 + 4 = 24$ (factors of 2 are in excess, so 5s are the bottleneck).

Concept Tested: Trailing zeros in a factorial

Question 8

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Convert decimal 156 to base 8. (Enter the base-8 digits, e.g., 234)

Answer / Expected Answer: 234

Solution: $156 \div 8 = 19 \text{ r}4$; $19 \div 8 = 2 \text{ r}3$; $2 \div 8 = 0 \text{ r}2$. Reading remainders bottom-up: 234₈.

Concept Tested: Base conversion

Question 9

Type: MCQ Source: PYQ-PATTERN Difficulty: Medium

Question: Which of the following is a prime number?

Options: (A) 91 (B) 97 (C) 111 (D) 133

Answer: (B)

Solution: $91 = 7 \times 13$, $111 = 3 \times 37$, $133 = 7 \times 19$ — all composite. 97 has no factor up to $\sqrt{97} \approx 9.8$, so it is prime.

Concept Tested: Primality checking

Question 10

Type: NAT Source: PYQ-PATTERN Difficulty: Medium

Question: Find the smallest positive number which, when divided by 4, 5, and 6, always leaves remainder 3.

Answer / Expected Answer: 63

Solution: $\text{LCM}(4,5,6) = 60$. Required number = $60 + 3 = 63$.

Concept Tested: LCM applied to a common-remainder problem

Question 11 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: Find the number of factors of 100.

Answer / Expected Answer: 9

Solution: $100 = 2^2 \times 5^2$. Number of factors = $(2+1)(2+1) = 9$.

Concept Tested: Number of factors formula

Question 12 (Original)

Type: NAT Source: ORIGINAL Difficulty: Easy

Question: Find HCF(48, 60).

Answer / Expected Answer: 12

Solution: $48 = 2^4 \times 3$, $60 = 2^2 \times 3 \times 5$. $\text{HCF} = 2^2 \times 3 = 12$.

Concept Tested: HCF from prime factorization

Question 13 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Find the remainder when 3^{40} is divided by 7.

Answer / Expected Answer: 4

Solution: Cycle of $3 \bmod 7$: 3, 2, 6, 4, 5, 1 (period 6). $40 \bmod 6 = 4 \rightarrow 4\text{th position} \rightarrow \text{remainder } 4$.

Concept Tested: Cyclicity in modular arithmetic

Question 14 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Find the last two digits of 7^{45} .

Answer / Expected Answer: 07

Solution: Last-two-digit cycle of 7: 07,49,43,01 (period 4). $45 \bmod 4 = 1 \rightarrow 1\text{st position} \rightarrow 07$.

Concept Tested: Cyclicity of last two digits

Question 15 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Find the sum of all factors of 96.

Answer / Expected Answer: 252

Solution: $96 = 2^5 \times 3$. Sum = $(2^0 + 2^1 + \dots + 2^5)(3^0 + 3^1) = 63 \times 4 = 252$.

Concept Tested: Sum of factors formula

Question 16 (Original)

Type: NAT Source: ORIGINAL Difficulty: Medium

Question: Find the number of trailing zeros in $50!$.

Answer / Expected Answer: 12

Solution: $\lfloor 50/5 \rfloor + \lfloor 50/25 \rfloor = 10 + 2 = 12$.

Concept Tested: Trailing zeros formula

Question 17 (Original)

Type: MCQ Source: ORIGINAL Difficulty: Medium

Question: Which of the following numbers is divisible by 11?

Options: (A) 2725 (B) 2726 (C) 2727 (D) 2728

Answer: (D)

Solution: Divisibility by 11: alternating digit sum. For 2728: $8 - 2 + 7 - 2 = 11$, divisible by 11. The others give non-multiples of 11.

Concept Tested: Divisibility rule for 11

Question 18 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: Find the remainder when 2^{100} is divided by 13.

Answer / Expected Answer: 3

Solution: By Fermat's Little Theorem, $2^{12} \equiv 1 \pmod{13}$. $100 \bmod 12 = 4$. So $2^{100} \equiv 2^4 = 16 \equiv 3 \pmod{13}$.

Concept Tested: Fermat's Little Theorem for modular exponentiation

Question 19 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: Find the number of positive integers less than 100 that are divisible by none of 2, 3, or 5.

Answer / Expected Answer: 26

Solution: Using inclusion-exclusion on 1-99: divisible by 2:49, by 3:33, by 5:19, by 6:16, by 10:9, by 15:6, by 30:3. Union = $49 + 33 + 19 - 16 - 9 - 6 + 3 = 73$. Neither = $99 - 73 = 26$.

Concept Tested: Inclusion-exclusion in number theory

Question 20 (Original)

Type: NAT Source: ORIGINAL Difficulty: Hard

Question: Find the smallest positive integer n such that $n!$ has at least 20 trailing zeros.

Answer / Expected Answer: 85

Solution: Trailing zeros(n) = $\lfloor n/5 \rfloor + \lfloor n/25 \rfloor + \dots$ For $n=84$: $16+3=19$ (insufficient). For $n=85$: $17+3=20$ (sufficient). So smallest $n = 85$.

Concept Tested: Inverse trailing-zeros problem

Answer Key — Number Systems

1-1 | 2-24 | 3-56 | 4-C | 5-3 | 6-4 | 7-24 | 8-234 | 9-B | 10-63 | 11-9 | 12-12 | 13-4 | 14-07 | 15-252 | 16-12 | 17-D | 18-3 | 19-26 | 20-85

COMPLETE ANSWER KEY — ALL 120 QUESTIONS

Standard Distributions: 1-C, 2-16/3, 3-B, 4-0.0498, 5-C, 6-45, 7-C, 8-1/9, 9-ABCD, 10-B, 11-B, 12-0.4, 13-B, 14-0.8647, 15-0.667, 16-0.0556, 17-ABC, 18-1.0, 19-0.3370, 20-B

Normal Distribution: 1-1.0, 2-C, 3-0, 4-B, 5-0.1587, 6-ABCD, 7-ABC, 8-16, 9-C, 10-4.21, 11-2.0, 12-B, 13-0.8413, 14-B, 15-62.82, 16-C, 17-ABC, 18-0.1336, 19-A, 20-15.79

Exponential Distribution: 1-0.5, 2-0.25, 3-C, 4-0.3679, 5-B, 6-0.8647, 7-ABD, 8-0.2, 9-C, 10-0.1733, 11-0.6321, 12-B, 13-0.3834, 14-0.2, 15-0.3679, 16-B, 17-0.2303, 18-0.5, 19-A, 20-0.1353

Joint Distribution: 1-B, 2-0.9, 3-0.5833, 4-0.09375, 5-(-0.02), 6-B, 7-0.3, 8-0.5714, 9-ACD, 10-B, 11-1.7, 12-0.5, 13-2.6, 14-0.05, 15-B, 16-0.5, 17-0.25, 18-0.5333, 19-0.4375, 20-20.2

Naive Bayes Classifier: 1-C, 2-0.6863, 3-0.2667, 4-B, 5-AB, 6-(-0.6778), 7-C, 8-19, 9-0.48, 10-B, 11-0.24, 12-A, 13-0.6667, 14-0.252, 15-0.036, 16-0.875, 17-ABD, 18-0.0417, 19-0.6899, 20-B

Number Systems: 1-1, 2-24, 3-56, 4-C, 5-3, 6-4, 7-24, 8-234, 9-B, 10-63, 11-9, 12-12, 13-4, 14-07, 15-252, 16-12, 17-D, 18-3, 19-26, 20-85